

Statistics of Financial Markets

Seminar 10: VaR, ES and backtesting



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **BET**: Bucharest Exchange Trading (index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **CF**: Cornish–Fisher (expansion)
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **EVT**: Extreme Value Theory
- **FHS**: Filtered Historical Simulation
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GPD**: Generalised Pareto Distribution
- **HS**: Historical Simulation
- **LR**: Likelihood Ratio (test)
- **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- **POF**: Proportion Of Failures (Kupiec test)
- **POT**: Peaks Over Threshold

Today's Question and Seminar Structure

- Question of the seminar
  - **how large can tomorrow's loss be, and how do we check a risk number afterwards?**
- Outputs to produce
  - the answers to the A exercises
  - the table or figure each B exercise asks for
  - a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Nothing is handed in: the seminar is for practice; the solutions of the [Proposed] exercises are discussed in class
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the Exercise Map (next slide)
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	VaR 1% and ES 2.5% of a position: Normal and Student-t	Solved, Proposed	A1
A3, A4	historical simulation from the worst days of the BET; a subadditivity counterexample	Solved, Proposed	A3
A5, A6	the Kupiec test and the traffic light; the Christoffersen test	Solved, Proposed	A5
B1, B2	five estimation methods: BET; TLV, SNP, Bitcoin	Solved, Proposed	B1
B3, B4	backtests per calendar year: S&P 500; BET, Bitcoin	Solved, Proposed	B3
B5, B6	two-asset portfolios: TLV and SNP; DAX and S&P 500	Solved, Proposed	B5
C1, C2	is ES 2.5% as large as VaR 1% on real data? what is wrong in an AI answer?	Proposed	B1, A5

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it after the model; the solution is discussed in the seminar
- A: calculations on paper; B: real data, in the notebook, each ending with an interpretation question; C: open questions, among them the audit of an AI answer

Data Used

Series	Source	Price used	Period
S&P 500, DAX, BET	EODHD	close	2000–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026
Banca Transilvania (TLV), OMV Petrom (SNP), BRD	EODHD	adjusted close	2010–2026

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; for portfolios, simple returns $R_t = 100(P_t/P_{t-1} - 1)$ on the common days; the last day is 18 September 2026
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, Chapter 2)
- In the notebook: `returns('bet')`, `hs_var(x)`, `kupiec(x, n)` (package `arch` for GARCH); no account or key is needed

Prerequisites for Today (1/21): Quantiles and Value at Risk

- X: the daily return (P&L, profit and loss) of a position, in %; level α: the probability of the left tail (1% or 2.5%)
  - α-**quantile** q_α: the value that X falls below with probability α

$$P(X \leq q_\alpha) = \alpha$$

  - for a small α, q_α is a negative return
- **VaR** (value at risk): the loss exceeded with probability α

$$\mathrm{VaR}_\alpha = -q_\alpha$$

  - a positive number; we write **VaR 1%**: "VaR 1% = 3%" means a loss of more than 3% on one day in a hundred
  - in money: $W \times \mathrm{VaR}/100$ for a position of value W
  - VaR says how often a large loss happens, not how large it is when it happens

Prerequisites for Today (2/21): Expected Shortfall

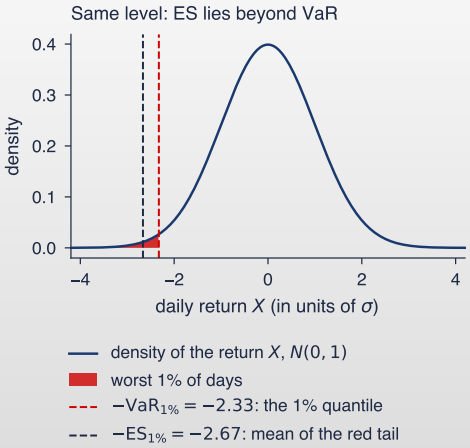
- **ES** (expected shortfall): the average loss on the worst α of days

$$\mathrm{ES}_\alpha = -E\left[X \mid X \le q_\alpha\right]$$

  - $E[X \mid X \le q_\alpha]$: the mean of X over the days in the tail only
  - we write **ES 2.5%**; Basel uses ES 2.5% for market risk BCBS (2019)
- ES and VaR at the **same** level

$$\mathrm{ES}_\alpha \ge \mathrm{VaR}_\alpha$$

  - the tail mean lies beyond the tail boundary; at different levels (ES 2.5%, VaR 1%) either can be larger
  - ratio ES 2.5% / VaR 1%: compares the two rules on the same data

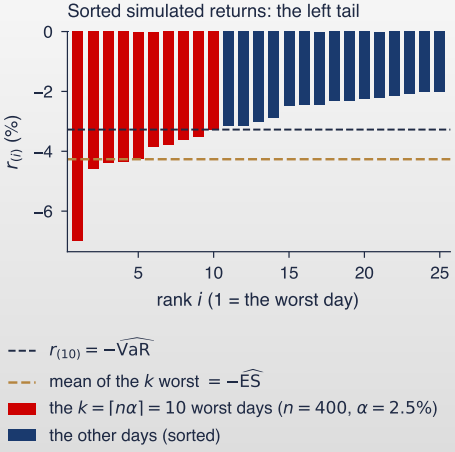


Prerequisites for Today (3/21): Historical Simulation

- **Historical simulation** (HS): the past n returns are taken as the possible outcomes of tomorrow
  - sort them: $r_{(\mathbf{1})} \le r_{(\mathbf{2})} \le \cdots \le r_{(n)}$; $r_{(i)}$: the i-th smallest return (an order statistic)
  - $k = \lceil n\alpha \rceil$: the number of days in the tail; $\lceil \cdot \rceil$: rounding up
- Estimates

$$\widehat{\mathrm{VaR}}_\alpha = -r_{(k)}, \qquad \widehat{\mathrm{ES}}_\alpha = -\frac{1}{k}\sum_{i=\mathbf{1}}^{k} r_{(i)}$$

  - VaR: the k-th worst day; ES: the mean of the k worst days
  - no distribution is assumed, but only losses already seen in the window can appear
  - **rolling** HS: the last 250 days, moved forward one day at a time



Prerequisites for Today (4/21): Normal VaR and ES

- Returns $X \sim N(\mu, \sigma^2)$: write $X = \mu + \sigma Z$, with $Z \sim N(0,1)$

$$\mathrm{VaR}_\alpha = -(\mu + \sigma z_\alpha), \qquad \mathrm{ES}_\alpha = -\mu + \sigma \frac{\varphi(z_\alpha)}{\alpha}$$

  - μ, σ: the mean and the standard deviation of the daily return
  - $z_\alpha = \Phi^{-1}(\alpha)$: the α-quantile of $N(0,1)$; Φ: the $N(0,1)$ distribution function
  - $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$: the $N(0,1)$ density; it comes from $E[Z \mid Z \le z_\alpha] = -\varphi(z_\alpha)/\alpha$
  - values: $z_{0.01} = -2.326$, $z_{0.025} = -1.960$, $\varphi(-1.960) = 0.0584$

- **Square-root-of-time rule** for h days (i.i.d. returns, mean 0)

$$\mathrm{VaR}^{(h)} = \sqrt{h}\,\mathrm{VaR}^{(1)}$$

  - the variance of a sum of h independent returns is $h\sigma^2$, so its standard deviation is $\sqrt{h}\,\sigma$

Prerequisites for Today (5/21): Student-t VaR and ES

- **Standardised Student-t** innovations z with $\nu > 2$ degrees of freedom (variance 1)
  - t_ν has variance $\nu/(\nu-2)$, so $z = t_\nu \sqrt{(\nu-2)/\nu}$ has variance 1; a small ν means heavy tails
  - $t_\alpha = t_\nu^{-1}(\alpha)$: the α-quantile of t_ν; f_ν: the density of t_ν
- Quantile and ES of the standardised innovation

$$q_\alpha(z) = t_\alpha \sqrt{\frac{\nu-2}{\nu}}, \qquad \mathrm{ES}_\alpha(z) = \sqrt{\frac{\nu-2}{\nu}}\,\frac{f_\nu(t_\alpha)}{\alpha}\,\frac{\nu + t_\alpha^2}{\nu-1}$$

  - the factor $(\nu + t_\alpha^2)/(\nu-1)$ makes ES grow faster than VaR when the tails are heavy
- For the return $X = \mu + \sigma z$

$$\mathrm{VaR}_\alpha = -\big(\mu + \sigma\, q_\alpha(z)\big), \qquad \mathrm{ES}_\alpha = -\mu + \sigma\, \mathrm{ES}_\alpha(z)$$

  - μ, σ, ν are estimated by maximum likelihood (Chapter 9)

Prerequisites for Today (6/21): Skewness, Kurtosis and the Cornish–Fisher Quantile

- **Skewness** S and **excess kurtosis** K of the standardised returns $z_t = (r_t - \bar{r})/\hat{\sigma}$

$$S = \frac{1}{n}\sum_{t=1}^{n} z_t^{3}, \qquad K = \frac{1}{n}\sum_{t=1}^{n} z_t^{4} - 3$$

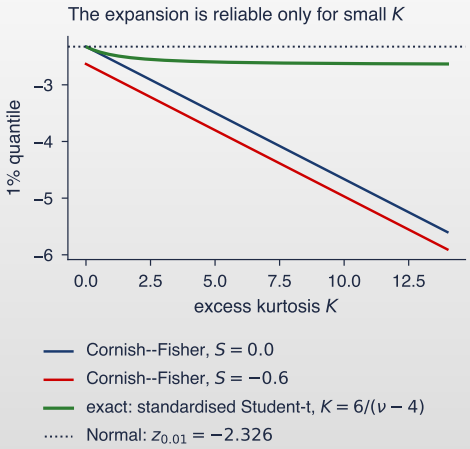
  - $S < 0$: the left tail is longer (large falls); $K > 0$: heavier tails than the Normal; both are 0 for the Normal

- **Cornish–Fisher** Cornish & Fisher (1938): the Normal quantile $z = z_\alpha$ corrected with S and K

$$z_{CF} = z + \frac{(z^2 - 1)\,S}{6} + \frac{(z^3 - 3z)\,K}{24} - \frac{(2z^3 - 5z)\,S^2}{36}$$

  - $\mathrm{VaR}_\alpha = -(\mu + \sigma z_{CF})$; for $z = -2.326$: $S < 0$ and $K > 0$ move z_{CF} further into the left tail
  - it is a series expansion around the Normal: accurate only when S and K are small (next slide)

Prerequisites for Today (7/21): The Cornish–Fisher Quantile and Heavy Tails



- The 1% quantile as a function of the excess kurtosis K
  - the Cornish–Fisher term $(z^3 - 3z)K/24$ is linear in K, so z_{CF} keeps falling as K grows
  - green: the exact 1% quantile of a standardised Student-t, whose excess kurtosis is $K = 6/(\nu - 4)$ for $\nu > 4$
- Reading
  - for small K the two curves are close; for large K the expansion moves far away from the exact quantile
  - a negative skewness S shifts the whole curve down

Prerequisites for Today (8/21): Extreme Value Theory: Peaks over Threshold

- **EVT** (extreme value theory), **POT** (peaks over threshold, Chapter 5): model only the losses $L = -X$ above a high threshold u
  - the excesses $y = L - u > 0$ follow approximately a **GPD** (generalised Pareto distribution)

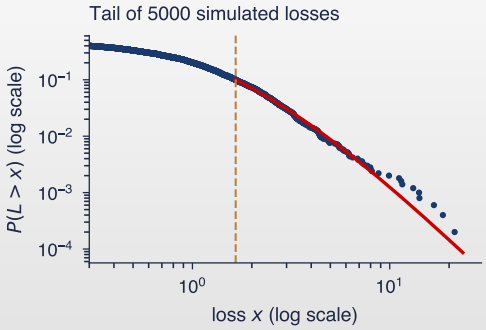
$$G_{\xi,\beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}$$

  - ξ: the shape (tail index), $\xi > 0$ for heavy tails; $\beta > 0$: the scale; both estimated by maximum likelihood
- **VaR and ES** from the fitted tail

$$\mathrm{VaR}_\alpha = u + \frac{\beta}{\xi}\left[\left(\frac{n\alpha}{N_u}\right)^{-\xi} - 1\right], \qquad \mathrm{ES}_\alpha = \frac{\mathrm{VaR}_\alpha + \beta - \xi u}{1 - \xi}$$

  - n: the number of days; N_u: the number of losses above u; N_u/n: the empirical probability of exceeding u
  - the ES formula needs $\xi < 1$; u is often the 90% quantile of the losses

Prerequisites for Today (9/21): A GPD Fitted to a Tail



- 5000 losses simulated from a Student-t with 3 degrees of freedom
- Log-log scale: the probability of a loss larger than x
  - above the threshold the GPD (red) follows the empirical tail; it also extrapolates beyond the largest observed losses
  - for this distribution the true $\xi = 1/3$; the estimate is close
  - a heavy tail appears as a straight line on this scale

Prerequisites for Today (10/21): Coherent Risk Measures

- A risk measure ρ maps a loss L to the capital $\rho(L)$ that makes the position acceptable; it is **coherent** Artzner et al. (1999) if, for all losses L, L_1, L_2:
  - **monotone**: $L_1 \leq L_2$ always $\Rightarrow \rho(L_1) \leq \rho(L_2)$
  - **translation invariant**: $\rho(L + c) = \rho(L) + c$ for a sure amount c
  - **positively homogeneous**: $\rho(\lambda L) = \lambda\, \rho(L)$ for $\lambda > 0$ (twice the position, twice the risk)
  - **subadditive**: diversification never adds risk

$$\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$$

- ES is coherent Acerbi & Tasche (2002); VaR is not always subadditive
  - the sum of the VaRs of the parts can be smaller than the VaR of the whole, typically for losses with jumps (defaults)

Prerequisites for Today (11/21): VaR and ES of a Discrete Loss

- A loss L with a few possible values (for example a bond that defaults or not)
  - **VaR**: the smallest loss level l that is exceeded with probability at most α

$$\mathrm{VaR}_\alpha = \min\{\, l : P(L > l) \le \alpha \,\}$$

  - **ES**: the average loss over the worst α of the probability mass
  - when a possible value (an atom) straddles the boundary, only the part of its probability that is needed to complete α is counted
- Two **independent** events, each with probability p

$$P(\text{both}) = p^2, \qquad P(\text{exactly one}) = 2p(1-p), \qquad P(\text{none}) = (1-p)^2$$

  - independence: the probability that both happen is the product of the probabilities

Prerequisites for Today (12/21): Backtesting: Exceptions and the Binomial Distribution

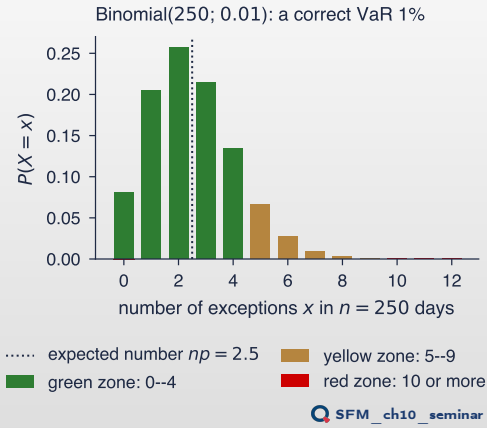
- **Exception**: a day with $r_t < -\mathrm{VaR}_t$, with the VaR forecast made the day before

$$I_t = 1\{r_t < -\mathrm{VaR}_t\}$$

  - $1\{\cdot\}$: 1 if the condition holds, 0 otherwise
- A correct VaR 1%: the I_t are i.i.d. Bernoulli(0.01)
  - the number x of exceptions in n days is Binomial(n, p), $p = 0.01$:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad E(X) = np$$

  - $\binom{n}{x}$: the number of ways to choose the x exception days among the n days
  - $P(X \le x)$: the cumulative probability, used for the Basel zones



Prerequisites for Today (13/21): The Kupiec Test

- **Kupiec** Kupiec (1995) (proportion of failures, POF): is the exception rate equal to $p = 1\%$?
  - x exceptions in n days; observed rate $\hat{\pi} = x/n$
- Binomial log-likelihoods with the rate p and with the observed rate $\hat{\pi}$

$$\ln L_0 = (n-x)\ln(1-p) + x \ln p, \qquad \ln L_1 = (n-x)\ln(1-\hat{\pi}) + x \ln \hat{\pi}$$

  - $\ln L_1 \geq \ln L_0$ always: $\hat{\pi}$ is the rate that fits the data best
- **Likelihood ratio** statistic and decision

$$LR_{uc} = -2\left[\ln L_0 - \ln L_1\right] \sim \chi^2(1) \text{ under } H_0 \colon \pi = p$$

  - uc: unconditional coverage; reject at 5% if $LR_{uc} > 3.84$ (p-value < 0.05)
  - too many **and** too few exceptions are rejected; with few days the test has little power

Prerequisites for Today (14/21): The Christoffersen Test

- **Christoffersen** Christoffersen (1998): do the exceptions come independently, or in clusters?
  - n_{ij}: the number of days with $I_{t-1} = i$ and $I_t = j$ $(i, j \in \{0, 1\})$

$$\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}, \qquad \hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}, \qquad \hat{\pi} = \frac{n_{01} + n_{11}}{n_{00} + n_{01} + n_{10} + n_{11}}$$

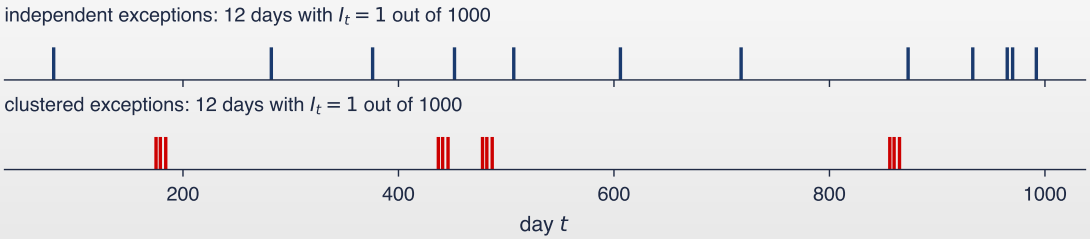
  - $\hat{\pi}_{01}$: the exception rate after a quiet day; $\hat{\pi}_{11}$: after an exception day
- Log-likelihoods without and with dependence on yesterday

$$\ln L(\hat{\pi}) = (n_{00} + n_{10}) \ln(1 - \hat{\pi}) + (n_{01} + n_{11}) \ln \hat{\pi}$$

$$\ln L(\hat{\pi}_{01}, \hat{\pi}_{11}) = n_{00} \ln(1 - \hat{\pi}_{01}) + n_{01} \ln \hat{\pi}_{01} + n_{10} \ln(1 - \hat{\pi}_{11}) + n_{11} \ln \hat{\pi}_{11}$$

- Statistics
  - $LR_{ind} = -2 \left[\ln L(\hat{\pi}) - \ln L(\hat{\pi}_{01}, \hat{\pi}_{11})\right] \sim \chi^2(1)$: independence; 5% critical value 3.84
  - $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$: conditional coverage (right rate **and** independence); 5% critical value 5.99

Prerequisites for Today (15/21): Independent and Clustered Exceptions



- Two simulated sequences with the same number of exceptions: the Kupiec test cannot tell them apart
- In the clustered sequence an exception is often followed by another one within a few days: the VaR adapts too slowly after a shock
- Christoffersen compares $\hat{\pi}_{\mathbf{11}}$ with $\hat{\pi}_{\mathbf{01}}$; with a correct VaR both are close to 1%

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Prerequisites for Today (16/21): The Basel Traffic Light and VaR Forecasts

- **Basel traffic light** BCBS (1996b): exceptions of VaR 1% in the last 250 days
  - green 0–4, yellow 5–9, red 10 or more; the zones follow the cumulative Binomial(250; 0.01) probabilities
  - capital = multiplier × average VaR; multiplier = 3+ plus factor
  - plus factors for 5, 6, 7, 8, 9 exceptions: 0.40, 0.50, 0.65, 0.75, 0.85; red zone: 1; green: 0
- VaR forecasts to backtest
  - **HS 250**: historical simulation on the last 250 days
  - **GARCH-t** (Chapter 9): $\mathrm{VaR}_{t+\mathbf{1}} = -(\hat{\mu} + \hat{\sigma}_{t+\mathbf{1}}\, q_{\mathbf{0.01}}(z))$, with $q_{\mathbf{0.01}}(z)$ the standardised Student-t quantile
  - **FHS** (filtered HS): HS on the standardised residuals $\hat{z}_t$, rescaled by $\hat{\sigma}_{t+\mathbf{1}}$
  - forecasts use only the data up to the previous day (out of sample)

Prerequisites for Today (17/21): Portfolio VaR and Diversification

- **Simple returns** add up across assets: $R_t = 100\,(P_t/P_{t-1} - 1)$

$$R_p = w_1 R_1 + w_2 R_2, \qquad \sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1 w_2 \rho\, \sigma_1\sigma_2$$

  - w_1, w_2: the weights ($w_1 + w_2 = 1$); σ_1, σ_2: the standard deviations; ρ: the correlation of the two returns

- **Variance–covariance** VaR (Normal)

$$\mathrm{VaR}_p = -(\mu_p + z_\alpha \sigma_p), \qquad \mu_p = w_1\mu_1 + w_2\mu_2$$

- **Diversification benefit**

$$1 - \frac{\mathrm{VaR}_p}{w_1\mathrm{VaR}_1 + w_2\mathrm{VaR}_2}$$

  - the share of the risk removed by combining the assets; 0 if $\rho = 1$, larger when ρ is smaller

Prerequisites for Today (18/21): Measuring Dependence in the Tails

- **Pseudo-observations**: the ranks of each series divided by $n+1$

$$U_{i,t} = \frac{\text{rank}(R_{i,t})}{n+1} \in (0,1)$$

  - $U_{i,t}$ close to 0: one of the worst days of asset i; the ranks remove the effect of the margins

- **Kendall's** τ: a rank correlation

$$\tau = \frac{\text{concordant pairs} - \text{discordant pairs}}{n(n-1)/2} \in [-1,1]$$

  - a pair of days is concordant if both assets move in the same direction from one day to the other

- **Tail dependence** at a small level q

$$\hat{\lambda}(q) = P(U_1 \leq q \mid U_2 \leq q) = \frac{\#\{t : U_{1,t} \leq q,\ U_{2,t} \leq q\}}{\#\{t : U_{2,t} \leq q\}}$$

  - $\#\{\cdot\}$: the number of days; under independence $\hat{\lambda}(q) \approx q$, and the expected number of joint days is nq^2
  - the correlation alone does not determine the tail dependence

Prerequisites for Today (19/21): Copulas

- **Copula** C: the joint distribution of (U_1, U_2); **Sklar**: any joint distribution splits into margins and a copula

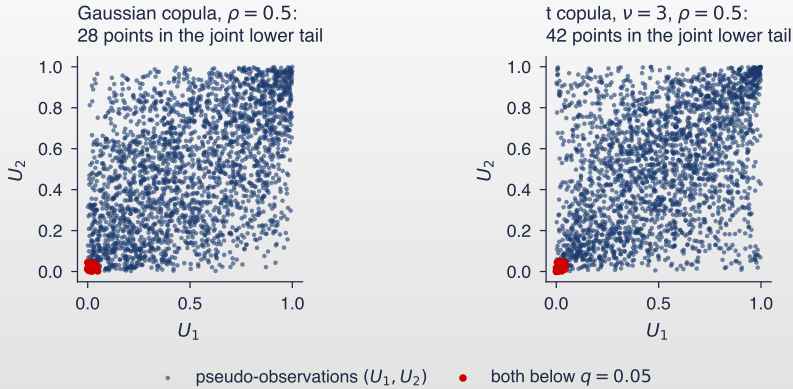
$$F(x_1, x_2) = C\big(F_1(x_1), F_2(x_2)\big)$$

  - F: the joint distribution function; F_1, F_2: the distribution functions of each asset (the margins)
- Two copulas with a correlation parameter ρ
  - **Gaussian copula**: no tail dependence ($\lambda = 0$ for $\rho < 1$)
  - **t copula** with ν degrees of freedom Demarta & McNeil (2005): positive tail dependence

$$\lambda = 2\, t_{\nu+1}\Big(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}}\Big) > 0$$

  - λ: the limit of $\hat{\lambda}(q)$ as $q \to 0$; $t_{\nu+1}(\cdot)$: the Student-t distribution function with $\nu + 1$ degrees of freedom
  - ρ from Kendall's τ: $\rho = \sin(\pi\tau/2)$; a small ν: strong tail dependence

Prerequisites for Today (20/21): Gaussian and t Copulas in Simulation



- Same ρ, 2500 simulated pairs: the t copula puts more points in the joint lower tail (both assets among their worst 5% of days)

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Prerequisites for Today (21/21): Copula VaR, Model Comparison and the Bootstrap

- **Copula VaR with empirical margins**
  - 1. fit the copula to the pseudo-observations; 2. simulate many pairs (U_1, U_2) from it
  - 3. turn them into returns with the empirical quantiles: $R_i = \hat{F}_i^{-1}(U_i)$; 4. compute R_p and its HS VaR and ES
- **Comparing two copulas**: $\mathrm{LR} = 2\,(\ell_t - \ell_G)$
  - ℓ_t, ℓ_G: the maximised log-likelihoods of the t and of the Gaussian copula; the t copula has one more parameter (ν)
  - a large LR: the extra parameter (tail dependence) improves the fit a lot
- **Bootstrap**: the uncertainty of a statistic by resampling
  - draw n days with replacement from the sample, recompute the statistic, repeat many times; the 2.5% and 97.5% quantiles of the results give a 95% interval

A1: VaR and ES of a Normal Position [Solved]

Task

- A position of 1 000 000 RON; daily returns $N(\mu, \sigma^2)$ with $\mu = 0.05\%$, $\sigma = 1.4\%$.
- 1. Compute VaR 1% in % and in RON.
- 2. Compute ES 2.5% in % and in RON.
- 3. Compute the 10-day VaR 1% with the square-root-of-time rule (μ ignored).
- 4. Compute the ratio ES 2.5% / VaR 1%.
- Report: six numbers and one sentence.

Solution

- 1. $-(0.05 - 2.326 \times 1.4) = -0.05 + 3.256 = 3.21\%$, i.e. 32 069 RON
- 2. $-0.05 + 1.4 \times 0.0584/0.025 = -0.05 + 3.270 = 3.22\%$, i.e. 32 229 RON
- 3. $\sqrt{10} \times 2.326 \times 1.4 = 10.30\%$, i.e. 102 992 RON
- 4. $3.22/3.21 = 1.005$
- For Normal returns ES 2.5% and VaR 1% are almost the same number.

Back Exercise list

A2: The Same Position with Student-t Returns [Proposed]

Solution discussed in class

Task

- As in A1, but the standardised returns are Student-t with $\nu = 4$: $X = \mu + \sigma z$; $t_4^{-1}(0.01) = -3.747$, $t_4^{-1}(0.025) = -2.776$, $f_4(-2.776) = 0.0256$. Model: A1.
- 1. Compute $q_{0.01}(z)$ and VaR 1% in % and in RON.
- 2. Compute $\mathrm{ES}_{2.5\%}(z)$ and ES 2.5% in % and in RON.
- 3. Compare both numbers with A1.
- Report: six numbers and one sentence.

A3: Historical Simulation from the Worst Days of the BET [Solved]

Task

- The last 500 daily BET returns (12 September 2024 – 18 September 2026); the 15 smallest, in %: -3.68; -3.35; -3.12; -3.03; -2.93; -2.71; -2.61; -2.41;
- -2.38; -2.33; -2.27; -2.11; -2.08; -2.02; -1.93.
- 1. Compute VaR 1% and VaR 2.5% by historical simulation.
- 2. Compute ES 2.5%.
- 3. Translate VaR 1% and ES 2.5% into RON for a position of 1 000 000 RON.
- Report: five numbers and one sentence.

Solution

- 1. $k = \lceil 500 \times 0.01 \rceil = 5$: VaR 1% $= 2.93\%$; $k = \lceil 12.5 \rceil = 13$: VaR 2.5% $= 2.08\%$
- 2. ES 2.5% $= 35.01/13 = 2.69\%$ (the average of the 13 smallest)
- 3. VaR 1%: 29 300 RON; ES 2.5%: 26 931 RON
- Here ES 2.5% is below VaR 1%: ES is larger than VaR only at the same level; ES 2.5% $\geq$ VaR 2.5% holds.

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A4: Two Bonds and Subadditivity [Proposed]

Solution discussed in class

Task

- Two independent bonds; each loses 100 with probability 3% and 0 otherwise; level 5%. Model: A3 and the prerequisite VaR and ES of a Discrete Loss (11/21).
- 1. Compute VaR 5% and ES 5% of one bond.
- 2. Write the distribution of the total loss of the two bonds.
- 3. Compute VaR 5% and ES 5% of the total loss.
- 4. Check subadditivity for VaR and for ES.
- Report: four numbers and two sentences.

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A5: The Kupiec Test and the Traffic Light [Solved]

Task

- A bank's VaR 1% was exceeded on 7 of the last 250 days. A second model was exceeded on 16 of 1000 days.
- 1. Compute $\ln L_0$, $\ln L_1$ and LR_{uc} for the bank, and decide at 5%.
- 2. Give the Basel zone, the plus factor and the capital multiplier.
- 3. Compute LR_{uc} for the second model and decide at 5%.
- Report: five numbers, two decisions and one sentence.

Solution

- 1. $243 \ln 0.99 + 7 \ln 0.01 = -34.68$; $243 \ln 0.972 + 7 \ln 0.028 = -31.93$; $LR_{uc} = 5.50 > 3.84$ (p $= 0.019$): rejected
- 2. Yellow zone ($P(X \leq 6) = 98.63\%$, $P(X \leq 7) = 99.60\%$); plus 0.65, multiplier 3.65
- 3. $984 \ln 0.99 + 16 \ln 0.01 = -83.57$; $984 \ln 0.984 + 16 \ln 0.016 = -82.03$; $LR_{uc} = 3.08$ (p $= 0.079$): not rejected
- A rate of 1.6% is not rejected with 1000 days; 2.8% is rejected with 250: the test needs a large deviation or many days.

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A6: The Christoffersen Test [Proposed]

Solution discussed in class

Task

- 1000 days of VaR 1% forecasts give the transition counts $n_{00} = 978$, $n_{01} = 8$, $n_{10} = 8$, $n_{11} = 5$. Model: A5.
- 1. Compute $\hat{\pi}_{01}$, $\hat{\pi}_{11}$ and $\hat{\pi}$.
- 2. Compute LR_{ind} and decide at 5%.
- 3. Compute LR_{uc} (with $x = 13$, $n = 999$) and LR_{cc}, and decide at 5%.
- Report: six numbers, three decisions and one sentence.

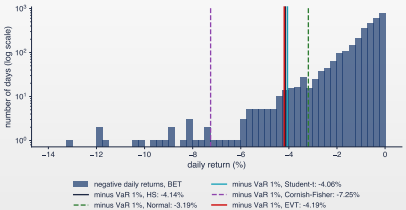
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B1: Five Methods for the BET [Solved]

- **Question**: how much do VaR 1% and ES 2.5% of the BET depend on the estimation method?
- **Inputs**: BET closes since 2000, daily log returns in %
- **Tasks**
  1. Compute VaR 1% and ES 2.5% by historical simulation, Normal and Student-t (maximum likelihood).
  2. Compute the skewness and the excess kurtosis, and the Cornish–Fisher VaR 1%.
  3. Fit a GPD to the losses above their 90% quantile and compute the EVT VaR 1% and ES 2.5%.
  4. Draw the left tail with the five values of minus VaR 1%.
  5. Interpretation: which number would you report to a risk committee?
- **Report**: a table of ten numbers, the chart and two sentences
- Notebook: section B1

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B1: Solution [Solved]



	HS	Normal	Student-t	Cornish–Fisher	EVT
VaR 1%	4.14	3.19	4.06	7.25	4.19
ES 2.5%	4.57	3.21	4.76	7.93	4.57

- $n = 6\,670$; $\hat{\mu} = 0.064$, $\hat{\sigma} = 1.40$; $S = -0.65$, $K = 11.1$; $\hat{\nu} = 2.43$; EVT: $u = 1.24\%$, $N_u = 667$, $\hat{\xi} = 0.23$
- The Normal VaR 1% is 23% below the data; Cornish–Fisher ($z_{CF} = -5.23$) is far above all others
- Interpretation: report the historical or the EVT values, which agree (VaR 1% about 4.14%, ES 2.5% about 4.57%); mention that the Normal understates and Cornish–Fisher is outside its range with $K > 10$

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B2: TLV, SNP and Bitcoin [Proposed]

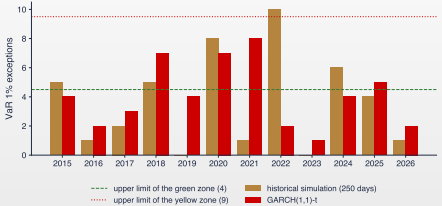
- **Question**: does the ranking of the methods found for the BET hold for two BVB stocks and for Bitcoin? Model: B1.
- **Inputs**: TLV and SNP since 2010 (TLV without 30–31 May 2016), Bitcoin since 2014; daily log returns in %
- **Tasks**
  1. Compute VaR 1% and ES 2.5% of each series by the five methods of B1.
  2. Compute by how much the Normal VaR 1% is below the historical VaR 1%.
  3. Report $\hat{\nu}$, $\hat{\xi}$ and the excess kurtosis of each series.
  4. Interpretation: for which series does the Normal understate VaR 1% most?
- **Report**: one table and two sentences
- Notebook: section B2

B3: Backtesting the S&P 500 Year by Year [Solved]

- **Question**: which of two VaR 1% models would a supervisor have penalised, year by year, since 2015?
- **Inputs**: S&P 500 since 2000; forecasts for every day since 2 January 2015
- **Tasks**
  1. Compute the one-day VaR 1% by historical simulation on the last 250 days.
  2. Compute the one-day GARCH(1,1)-t VaR 1%, re-estimating the model at the start of every year on all earlier data.
  3. Count the exceptions of each model in every calendar year and give the Basel zone of each year.
  4. Run the Kupiec and Christoffersen tests on the whole period.
  5. Interpretation: which model would a supervisor have penalised in 2022?
- **Report**: the chart, six statistics with p-values and two sentences
- Notebook: section B3

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B3: Solution [Solved]



	n	x	rate (%)	LR_{uc} (p)	LR_{ind} (p)	LR_{cc} (p)	red/yellow years
HS 250	2 944	43	1.46	5.52 (0.019)	23.04 (<0.001)	28.56 (<0.001)	1/4
GARCH-t	2 944	49	1.66	10.94 (<0.001)	6.78 (0.009)	17.72 (<0.001)	0/4

- Both models are exceeded too often (expected about 29.4); HS exceptions cluster strongly ($n_{\mathbf{11}} = 7$, p ind. < 0.001), GARCH-t less ($n_{\mathbf{11}} = 4$)
- Interpretation: in 2022 HS had 10 exceptions (red zone), GARCH-t 2: the slow window was penalised when volatility rose; GARCH-t paid in calmer years (2021: 8)

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B4: Backtesting the BET and Bitcoin [Proposed]

- **Question**: does the conclusion of B3 hold on the BVB and for Bitcoin? Model: B3.
- **Inputs**: BET since 2000, forecasts since 2015; Bitcoin since 2014, forecasts since 2016
- **Tasks**
  1. Compute both VaR 1% forecasts as in B3.
  2. Count the exceptions per calendar year and give the zones.
  3. Run the Kupiec and Christoffersen tests on the whole period.
  4. Interpretation: which model would you keep for each series?
- **Report**: one table and two sentences
- Notebook: section B4

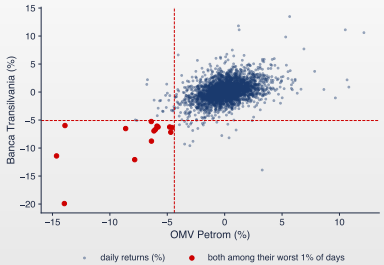
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B5: A Portfolio of Two BVB Stocks [Solved]

- **Question**: how much does diversification between Banca Transilvania and OMV Petrom reduce VaR 1%, and does it survive a crisis?
- **Inputs**: TLV and SNP adjusted closes since 2010, joined on common days; simple returns in %; weights 50/50
- **Tasks**
  1. Compute the variance–covariance VaR 1% of the portfolio, the two individual VaRs and the diversification benefit.
  2. Compute the historical VaR 1% and ES 2.5% of the portfolio.
  3. Compute the correlation in 2017 and in February–May 2020, and the tail dependence at $q = 5\%$ and $q = 1\%$.
  4. Fit a t copula and compute the copula VaR 1% with empirical margins.
  5. Interpretation: why does the variance–covariance VaR understate the risk of this portfolio?
- **Report**: eight numbers, the chart and two sentences
- Notebook: section B5

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B5: Solution [Solved]



- $n = 3\,461$; $\hat{\sigma}$: TLV 1.84, SNP 1.70; $\hat{\rho} = 0.46$; $\sigma_p = 1.511$; VaR 1%: TLV 4.17, SNP 3.85, portfolio 3.41%; benefit 15%
- Historical: VaR 1% = 4.12%, ES 2.5% = 4.51%; t copula ($\hat{\nu} = 5.00$, $\lambda = 0.166$): VaR 1% = 3.96%
- Correlation: 2017 0.32, spring 2020 0.69; tail dependence 0.36 ($q = 5\%$), 0.40 ($q = 1\%$); joint worst-1% days: 14, against 0.35 under independence
- Interpretation: the Normal VaR is 17% below the historical one: both margins have heavy tails and the two stocks fall together on the worst days, when the correlation is highest

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B6: DAX and S&P 500 [Proposed]

- **Question**: is the Gaussian copula good enough for a 50/50 DAX and S&P 500 portfolio? Model: B5.
- **Inputs**: DAX and S&P 500 closes since 2000, joined on common days; simple returns in %
- **Tasks**
  1. Compute the variance–covariance and the historical VaR 1% and ES 2.5%.
  2. Fit the Gaussian and the t copula and compare them with the LR statistic.
  3. Compute the copula VaR 1% and ES 2.5% with empirical margins for both copulas.
  4. Interpretation: is the Gaussian copula good enough for this pair?
- **Report**: one table and two sentences
- Notebook: section B6

C1: Is ES 2.5% as Large as VaR 1%? [Proposed]

- **Question**: Basel replaced VaR 1% by ES 2.5% because the two are equal for Normal returns; how close are they on real data and in crises?
- **Inputs**: S&P 500, DAX, BET, Bitcoin, TLV, SNP, BRD; models: B1, A5
- **Tasks**
  1. Compute the ratio ES 2.5% / VaR 1% by historical simulation on the whole sample of each series.
  2. Compute the same ratio on the 500 days that end in March 2009 and in May 2020.
  3. Compute the ratio implied by a fitted Student-t distribution.
  4. Interpretation: is ES 2.5% a stricter rule than VaR 1% on these markets?
- **Report**: a table and a plan for a project
- Notebook: section C1

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C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to summarise the risk of the BET over the last 500 days. The answer:
- (a) The VaR 99% of the BET is −2.93%.
- (b) ES is always at least as large as VaR, so ES 2.5% is above 2.93%.
- (c) The 10-day VaR 1% is 10 x 2.93 = 29.3%.
- (d) With 8 exceptions in 250 days the Kupiec p-value is 0.005, but the model stays in the green zone.
- (e) VaR is subadditive, so the VaR of a portfolio never exceeds the sum of the VaRs of its parts.
- (f) If the exceptions cluster, the Christoffersen test can reject even when their number is right.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

Points to Remember

- VaR 1% $= -q_{1\%}$, a positive loss; ES 2.5% averages the worst 2.5% of days; ES $\geq$ VaR only at the same level
- On real data the Normal method understates VaR; Cornish–Fisher fails for large kurtosis; HS and EVT agree at 1%
- Backtesting needs two checks: the number of exceptions (Kupiec) and their independence (Christoffersen)
- Diversification is weakest in crises: correlation and tail dependence rise together
- An AI answer is a draft: check the sign and the level of VaR, the horizon rule and the Basel zone

After the Seminar

- Lecture 10 develops each topic of today: definitions and coherence, estimation methods, conditional VaR, portfolios and copulas, backtesting
- C1 can grow into a team project: rolling ratios, bootstrap intervals, the capital under each rule
- Reading: Franke, Härdle & Hafner (2019), Ch. 16; exercises in Borak, Härdle & López-Cabrera (2013), Ch. 16; McNeil, Frey & Embrechts (2015), Ch. 2

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